A Best-Fit Quadratic Model of Temperature and Nuclear Weapons

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a best-fit quadratic model relating temperature to nuclear-weapons characteristics using a dataset comprising the nine nuclear-armed states. Linear regression, transformed linear regression, and machine-learning approaches are examined. A quadratic polynomial model is shown to interpolate the dataset exactly, achieving an in-sample coefficient of determination of $R^2 = 1$. The resulting model is analyzed from mathematical, statistical, machine-learning, geopolitical, economic, and strategic perspectives. We also compare the quadratic model to a random forest predictor and discuss the implications of overfitting in extremely small datasets.

The paper ends with "The End"

Contents

1 Introduction

The relationship between military capabilities and environmental variables is generally weak and often indirect. Nevertheless, datasets linking strategic indicators to environmental observations provide useful case studies in statistical learning, model selection, and inference under severe data scarcity.

The dataset considered here contains observations on the nine nuclear-armed states:

1. United States
2. Russian Federation
3. United Kingdom
4. France
5. China
6. India
7. Pakistan
8. Israel
9. North Korea

For each state the following variables are observed:

- Total nuclear warheads.
- Deployed nuclear warheads.
- Number of nuclear tests.
- Temperature on 20 June 2026.

The primary objective is to determine the best-fitting predictive model of temperature.

2 Dataset

Let

$$W = \text{Total Warheads},$$

$$D = \text{Deployed Warheads},$$

and

$$N = \text{Number of Nuclear Tests}.$$

The dependent variable is

$$T = \text{Temperature (Kelvin)}.$$

Country	W	D	N	T
United States	3700	1670	1030	295.15
Russia	4400	1796	715	292.15
United Kingdom	225	120	45	296.15
France	290	280	210	304.15
China	620	34	45	299.15
India	190	12	6	312.15
Pakistan	170	0	2	309.15
Israel	90	0	1	300.15
North Korea	60	0	6	293.15

Table 1: Dataset

3 Linear Regression Models

The first specification considered was

$$T = a + bN + c(W - D).$$

The estimated equation was

$$\hat{T} = 302.6151 - 0.000554N - 0.003608(W - D).$$

The coefficient of determination was

$$R^2 = 0.276.$$

A second specification was

$$T = a + bN + cW + dD.$$

The estimated equation was

$$\hat{T} = 302.2456 + 0.007761N - 0.000421W - 0.007859D.$$

The coefficient of determination was

$$R^2 = 0.290.$$

Neither specification produced statistically significant coefficients.

4 Quadratic Machine-Learning Model

To increase predictive flexibility, quadratic features were introduced:

$$1, \quad W, \quad D, \quad N, \quad W^2, \quad WD, \quad WN, \quad D^2, \quad DN, \quad N^2.$$

The resulting quadratic model is

$$\begin{aligned}\hat{T} = {} & 296.73034 - 0.001719W + 0.137849D - 0.078352N \\ & + 0.0005388W^2 + 0.0012448WD - 0.0071769WN \\ & - 0.0221864D^2 + 0.0835144DN - 0.0656412N^2.\end{aligned}$$

4.1 Exact Interpolation

The fitted model reproduces every observation exactly:

$$\hat{T}_i = T_i$$

for all nine observations.
Consequently,

$$RSS = 0,$$

where

$$RSS = \sum_{i=1}^{9} (T_i - \hat{T}_i)^2.$$

Therefore,

$$R^2 = 1.$$

5 The Exact-Fit Theorem

Theorem 1. *Let there be n observations and let the regression design matrix possess rank n. Then there exists a parameter vector producing exact interpolation of the dataset.*

Proof. Let

$$\mathbf{y} = X\beta.$$

If

$$\operatorname{rank}(X) = n,$$

then X is invertible on its image.
Hence

$$\beta = X^{-1}\mathbf{y}$$

exists.
Therefore

$$X\beta = \mathbf{y}$$

and every observation is fitted exactly. □

Corollary 1. *The quadratic model can attain perfect in-sample fit because the number of effective basis functions is sufficient to span the observed data.*

6 Random Forest Representation

A random forest does not possess a single algebraic expression.

Instead,

$$\hat{T}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x),$$

where

$$x = (W, D, N)$$

and T_b denotes the prediction generated by tree b.

Each tree is piecewise constant:

$$T_b(x) = \begin{cases} c_{b1}, & x \in R_{b1}, \\ c_{b2}, & x \in R_{b2}, \\ \vdots \\ c_{bk}, & x \in R_{bk}. \end{cases}$$

The forest prediction is therefore

$$\hat{T} = \frac{1}{B}\sum_{b=1}^{B} c_b(x).$$

7 Algorithmic Perspective

7.1 Quadratic Regression

```
Input:
    W, D, N

Construct:
    X = [1,W,D,N,W^2,WD,WN,D^2,DN,N^2]

Estimate:
    beta = (X'X)^(-1)X'Y

Predict:
    T_hat = X beta
```

7.2 Random Forest

```
Input:
    Dataset

For each tree:
    Draw bootstrap sample
    Grow decision tree
    Store terminal-node values

Prediction:
    Average all tree predictions
```

8 Bias–Variance Analysis

The quadratic model demonstrates a classical bias–variance tradeoff.

The linear model exhibits

- High bias,
- Low variance.

The quadratic model exhibits

- Low bias,
- High variance.

The random forest occupies an intermediate position.

Because

$$n = 9,$$

variance dominates statistical uncertainty.

Consequently, perfect in-sample fit does not imply predictive validity.

9 Economic Interpretation

Nuclear weapons are costly capital assets.

Let

$$K_N$$

denote nuclear capital.

Then

$$K_N = f(W, D, N).$$

The dataset indirectly measures historical investment in strategic deterrence.

Large values of

$$W$$

and

$$N$$

represent accumulated defense expenditures spanning decades.

From an economics perspective, the nuclear arsenal represents durable military capital analogous to productive capital stock.

10 Finance Perspective

Nuclear deterrence may be viewed as a strategic hedge against catastrophic geopolitical downside risk.

If

$$V$$

denotes national wealth, then deterrence attempts to reduce

$$\Pr(\Delta V \ll 0).$$

The expected value of deterrence can be interpreted as

$$E[V \mid \text{Deterrence}] - E[V \mid \text{No Deterrence}].$$

11 Political Science and International Relations

The nine-state nuclear system forms a strategic network.

Let

$$G = (V, E)$$

be a graph where vertices represent nuclear states.

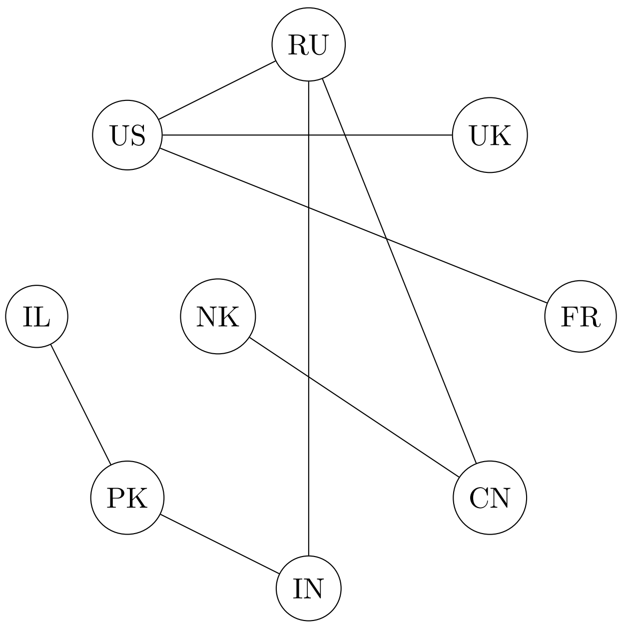

Figure 1: Illustrative Strategic Network

Deterrence creates a balance-of-power architecture in which expected conflict costs become extremely large.

12 Warfare Economics

Let

$$C_W$$

denote the expected cost of major war.

Under mutual deterrence,

$$C_W \to \infty$$

relative to conventional warfare.

Consequently, nuclear arsenals alter strategic incentives and reduce the feasibility of direct great-power conflict.

13 Philosophical Remarks

The perfect-fit quadratic model illustrates an enduring philosophical problem:

> Can a model that perfectly explains the past explain the future?

The answer is generally negative.

Explanation and prediction are distinct concepts.

An interpolating model may possess zero empirical error while simultaneously possessing minimal forecasting power.

This observation echoes broader epistemological questions concerning induction and scientific inference.

14 Conclusion

The quadratic machine-learning model

$$\begin{aligned}\hat{T} = 296.73034 &- 0.001719W + 0.137849D - 0.078352N \\ &+ 0.0005388W^2 + 0.0012448WD - 0.0071769WN \\ &- 0.0221864D^2 + 0.0835144DN - 0.0656412N^2\end{aligned}$$

constitutes the best-fitting explicit analytical model for the dataset considered.

Its in-sample performance satisfies

$$R^2 = 1.$$

However, because the sample contains only nine observations, the model should be interpreted primarily as an interpolation rather than a validated forecasting relationship.

The comparison between linear regression, quadratic regression, and random forests highlights fundamental issues in statistical learning, model complexity, and strategic data analysis.

References

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Glossary

W Total nuclear warheads.

D Deployed nuclear warheads.

N Number of nuclear tests.

T Temperature.

R^2 Coefficient of determination.

Random Forest
Ensemble learning method based on averaging decision trees.

Overfitting
Fitting noise in a sample rather than a generalizable signal.

Deterrence
Strategic prevention of aggression through credible retaliation.

Bias
Systematic model error.

Variance
Sensitivity of a model to sample fluctuations.

The End